POLYNOMIAL MINIMAL GRAPHS IN DIMENSION EIGHT ARE AFFINE

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ABSTRACT. We prove that every real scalar polynomial on $\mathbb{R}^8$ whose graph is a minimal hypersurface in $\mathbb{R}^9$ is affine, without a bound on its degree. The principal ingredient is a complex-algebraic obstruction: an irreducible homogeneous polynomial h satisfying $h \mid \mathcal{L}h$ and having no nonzero real zero must be a scalar multiple of the Euclidean null quadric. The proof maximizes $|z \cdot z|/|z|^2$ on a projective hypersurface and treats singular maximizers by estimates for the local holomorphic sheets. A radial-factor identity then excludes positive homogeneous divisors of nonzero homogeneous solutions of $\mathcal{L}F = 0$. Guo's structure theorem and strict degree estimate finish the argument in dimension eight. We also document a companion Lean proof of the same conclusion, which proceeds through calibrated homogeneous rigidity and dimensional induction.

1. INTRODUCTION

For a twice differentiable scalar function $P\colon \mathbb{R}^n \to \mathbb{R}$, its graph is minimal precisely when

$$\mathcal{M}P := (1 + |\nabla P|^2)\Delta P - \sum_{i,j=1}^{n} P_i P_j P_{ij} = 0. \tag{1.1}$$

The polynomial case retains the geometric content of this equation while allowing one to use the factorization of the highest homogeneous component. Guo's work [2] supplies a structure theorem for that component and a strict constraint on the degrees of its factors. Our purpose is to combine those results with a complex-algebraic obstruction to obtain the following conclusion.

Theorem 1.1. *Let $P \in \mathbb{R}[x_1, \dots, x_8]$. If $\mathcal{M}P = 0$, then there are $a \in \mathbb{R}$ and $b_1, \dots, b_8 \in \mathbb{R}$ such that*

$$P(x) = a + \sum_{i=1}^{8} b_i x_i.$$

Equivalently, every scalar real polynomial minimal graph over $\mathbb{R}^8$ is an affine hyperplane in $\mathbb{R}^9$.

The theorem concerns polynomials of arbitrary degree. Throughout the paper, "dimension eight" refers to the domain, not to the ambient space of the graph. The polynomial identity $\mathcal{M}P = 0$ is equivalent to its pointwise validity on $\mathbb{R}^8$.

We separate the proof into three steps. First, we classify homogeneous irreducible complex hypersurfaces that satisfy the minimal level-set equation and avoid the real projective locus. Second, we exclude the resulting radial factor and hence every positive homogeneous real divisor. Third, we apply Guo's structure theorem and degree estimate. The final section records the scope and organization of the companion formal proof of the theorem.

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1.1. **Notation.** For real or complex polynomials define

$$\mathcal{L}f = (\nabla f \cdot \nabla f)\Delta f - \sum_{i,j=1}^{n} f_i f_j f_{ij}, \qquad \mathcal{M}f = \Delta f + \mathcal{L}f. \tag{1.2}$$

Over $\mathbb{C}$ the product $z \cdot v = \sum_i z_i v_i$ is complex bilinear. It contains no complex conjugation. We reserve $|z|^2 = \sum_i z_i \overline{z_i}$ for the positive Hermitian norm and put

$$q(z) = z \cdot z = z_1^2 + \cdots + z_n^2.$$

All local bounds below use Hermitian norms and are uniform over the indicated roots, counted with multiplicity.

2. Two local analytic lemmas

The main obstruction is obtained at a maximum on a projective hypersurface. That maximum need not be a smooth point. The next two lemmas allow the argument to include a singular point without selecting a differentiable branch there.

Lemma 2.1 (Quadratic bounds for local sheets)**.** *Let*

$$W(\zeta, w) = w^s + a_1(\zeta)w^{s-1} + \cdots + a_s(\zeta), \qquad s \geq 1,$$

have holomorphic coefficients near $0 \in \mathbb{C}^r$, with $W(0, w) = w^s$. Let $w_j(\zeta)$ denote its roots. Suppose, uniformly in j, that

$$w_j = o(|\zeta|), \qquad \operatorname{Re}(\beta w_j) \leq C|\zeta|^2$$

for a fixed $\beta \in \mathbb{C} \setminus \{0\}$. Then $w_j = O(|\zeta|^2)$ uniformly in j.

Proof. The mean $\mu = s^{-1}\sum_j w_j = -a_1/s$ is holomorphic. Since $\mu = o(|\zeta|)$, its constant and linear Taylor coefficients vanish, and $\mu = O(|\zeta|^2)$. Set $\omega_j = \beta w_j$. Each $\operatorname{Re}\omega_j$ has the assumed upper bound, and their sum is $s\operatorname{Re}(\beta\mu) = O(|\zeta|^2)$. Subtracting the upper bounds for the other $s-1$ terms gives a lower bound of the same order. Consequently

$$|\operatorname{Re}\omega_j| \leq C_1|\zeta|^2. \tag{2.1}$$

Newton's identities show that $S_2 = \sum_j \omega_j^2$ is holomorphic. Moreover,

$$\operatorname{Re} S_2 = \sum_j (\operatorname{Re}\omega_j)^2 - \sum_j (\operatorname{Im}\omega_j)^2 \leq C_2|\zeta|^4.$$

This one-sided estimate forces every homogeneous Taylor term of S_2 of degree less than four to vanish. Indeed, if its first nonzero term were H_d with $d < 4$, choose v with $H_d(v) \neq 0$ and choose the phase of t so that $H_d(tv)$ is positive real. The resulting positive term of order $|t|^d$ contradicts the displayed upper bound. Thus $|S_2| \leq C_3|\zeta|^4$. Combining this with (2.1) gives

$$\sum_j (\operatorname{Im}\omega_j)^2 = \sum_j (\operatorname{Re}\omega_j)^2 - \operatorname{Re} S_2 \leq C_4|\zeta|^4.$$

Both real and imaginary parts of every ω_j are therefore $O(|\zeta|^2)$. Since $\beta \neq 0$, the assertion follows. □

Lemma 2.2 (Trace of the sheet mean)**.** *Let a reduced holomorphic hypersurface germ at the origin have equation $U(\zeta, w)W(\zeta, w) = 0$, where U is a holomorphic unit and W is as in Lemma 2.1. Suppose its tangent cone is $w = 0$ and its roots satisfy $w_j = O(|\zeta|^2)$. Equip the ambient coordinates with the constant complex bilinear metric*

$$\begin{pmatrix} G & 0 \\ 0 & c \end{pmatrix}, \qquad G = G^{\mathsf{T}} \text{ invertible}, \quad c \neq 0.$$

If the hypersurface satisfies $\mathcal{L}h = 0$ on its zero set, with the operator interpreted in this metric, then the holomorphic mean $\mu = s^{-1}\sum_j w_j$ satisfies

$$\Delta_G \mu(0) = 0, \qquad \Delta_G = \sum_{i,j} (G^{-1})_{ij}\partial_i\partial_j. \tag{2.2}$$

Proof. The discriminant $D(\zeta)$ of W is not identically zero: a reduced monic Weierstrass polynomial is squarefree over the field of meromorphic germs, which has characteristic zero. Let D_d be its first nonzero homogeneous Taylor term and choose v with $D_d(v) \neq 0$. For some $\varepsilon > 0$ and all sufficiently small positive real t, the ball $B(tv, \varepsilon t)$ contains no zero of D. In fact, choose ε so that D_d has no zero on $\overline{B(v, \varepsilon)}$ and use the uniform expansion $D(t\eta) = t^d(D_d(\eta) + O(t))$ there.

The roots of W can be labeled holomorphically on each such ball. The quadratic bound on the roots and Cauchy's estimates on a smaller concentric ball give, at $\zeta = tv$,

$$w_j = O(t^2), \qquad \partial w_j = O(t), \qquad \partial^2 w_j = O(1).$$

For a local sheet $w = f(\zeta)$ write

$$\nabla_G f = G^{-1}\partial f, \qquad A_f = (\partial f)^{\mathsf{T}} G^{-1}(\partial f).$$

The defining equation $w - f(\zeta)$ has squared normal length $c^{-1} + A_f$, Laplacian $-\Delta_G f$, and Hessian contraction $-D^2 f(\nabla_G f, \nabla_G f)$. Hence its level-set equation is

$$(1 + cA_f)\Delta_G f - cD^2 f(\nabla_G f, \nabla_G f) = 0. \tag{2.3}$$

Changing a local defining equation by a holomorphic unit multiplies its $\mathcal{L}$ value on the zero set by the cube of that unit. Thus (2.3) holds on each sheet.

At tv, we have $A_f = O(t^2)$ and the Hessian contraction in (2.3) is $O(t^2)$. Since $1 + cA_f$ stays nonzero for small t, it follows that $\Delta_G w_j(tv) = O(t^2)$. Averaging and using the holomorphic extension of μ across the discriminant locus yields $\Delta_G \mu(tv) = O(t^2)$. Letting $t \to 0$ proves (2.2). □

3. A complex hypersurface obstruction

Theorem 3.1. *Let $n \geq 3$ and let $h \in \mathbb{C}[z_1, \ldots, z_n]$ be a nonconstant irreducible homogeneous polynomial. Assume that $h \mid \mathcal{L}h$ and that $h(x) \neq 0$ for every $x \in \mathbb{R}^n \setminus \{0\}$. Then*

$$h(z) = \alpha q(z) \quad \text{for some } \alpha \in \mathbb{C} \setminus \{0\}.$$

Proof. Put $X = \{h = 0\}$ and consider

$$K(z) = \frac{|q(z)|}{|z|^2}, \qquad z \neq 0.$$

The function is invariant under nonzero complex scaling, so it attains a maximum κ on the nonempty compact projective hypersurface defined by h. The triangle inequality gives $0 \leq K \leq 1$. Equality holds exactly when z is a complex multiple of a nonzero real vector. The hypothesis on real zeros therefore implies $\kappa < 1$.

If $\kappa = 0$, then q vanishes on X. The complex Nullstellensatz gives $h \mid q$. Since q has rank $n \geq 3$, it is irreducible over $\mathbb{C}$ and the conclusion follows. Suppose, for a contradiction, that $0 < \kappa < 1$.

Choose a maximizing point z_0 and scale it so that $|z_0|^2 = 1$ and $q(z_0) = \kappa > 0$. Write $z_0 = x+iy$, with $x, y \in \mathbb{R}^n$. Then

$$x \cdot y = 0, \qquad |x|^2 = \frac{1+\kappa}{2}, \qquad |y|^2 = \frac{1-\kappa}{2}. \tag{3.1}$$

In particular, x and y are both nonzero. Define

$$N = z_0 - \kappa\overline{z_0}, \qquad c = N \cdot N = -\kappa(1-\kappa^2) \neq 0. \tag{3.2}$$

We have $N \cdot z_0 = 0$. The real-analytic function $\Phi = \log K$ is defined near z_0, and its holomorphic differential is

$$\partial\Phi(z_0)[v] = \Big(\frac{z_0}{\kappa} - \overline{z_0}\Big) \cdot v = \frac{N \cdot v}{\kappa}. \tag{3.3}$$

The tangent cone at the maximum. Let H_s be the first nonzero homogeneous Taylor term of $h(z_0 + v)$. Every $v \in \{H_s = 0\}$ is a limit of positive real rescalings of vectors in $X - z_0$. To see this directly, choose u with $H_s(u) \neq 0$. The polynomial $H_s(v+\tau u)$ is not identically zero and has a root at $\tau = 0$. For positive $t \to 0$,

$$t^{-s} h\big(z_0 + t(v + \tau u)\big) \longrightarrow H_s(v + \tau u)$$

uniformly on compact τ-disks. Applying Rouché's theorem on sufficiently small disks, and then taking a diagonal sequence of shrinking disks, gives $t_\nu \downarrow 0$ and $\tau_\nu \to 0$ with $z_0 + t_\nu(v + \tau_\nu u) \in X$. Conversely, Taylor expansion shows that any limit of rescaled zeros belongs to $\{H_s = 0\}$.

The maximum inequality and (3.3) imply $\operatorname{Re}(N \cdot v) \leq 0$ for every vector in this tangent cone. Applying the same statement to every complex multiple of v gives $N \cdot v = 0$. Thus the tangent cone lies in $T = N^\perp$. Since its equation is a homogeneous polynomial, unique factorization and the Nullstellensatz imply

$$H_s(v) = A(N \cdot v)^s \qquad (A \neq 0).$$

Indeed, the zero set of every irreducible factor of H_s lies in the hyperplane $N \cdot v = 0$, so that factor divides the linear polynomial $N \cdot v$.

Control of every local sheet. Because $N \cdot N \neq 0$, use local coordinates

$$z = z_0 + \zeta + wN, \qquad \zeta \in T. \tag{3.4}$$

Weierstrass preparation gives $h = UW$, where U is a holomorphic unit and W is monic of degree s in w, with $W(0, w) = w^s$. The tangent-cone calculation implies $w_j = o(|\zeta|)$ uniformly for all local roots. More explicitly, the lowest Taylor term in these coordinates is a nonzero multiple of w^s. If a sequence of zeros had $|w| \geq \varepsilon|\zeta|$, the Taylor equation would imply

$$|A'|\,|w|^s \leq C(|w| + |\zeta|)^{s+1}, \qquad A' \neq 0,$$

which is impossible as the sequence approaches the origin.

Expanding $\Phi \leq \Phi(z_0)$ along the zero set and using $N \cdot \zeta = 0$ gives

$$2\operatorname{Re}(cw_j/\kappa) \leq C(|\zeta|^2 + |w_j|^2).$$

The estimate $w_j = o(|\zeta|)$ makes the right-hand side $O(|\zeta|^2)$. Lemma 2.1, with $\beta = c$, now gives

$$w_j = O(|\zeta|^2). \tag{3.5}$$

Let $\mu = s^{-1}\sum_j w_j$ denote their holomorphic mean.

A vanishing trace. Set $E = \{v : x \cdot v = y \cdot v = 0\}$. Then $T = \mathbb{C}z_0 \perp E$ for the bilinear product. Choose a real orthonormal basis $e_1, \ldots, e_{n-2}$ of the real orthogonal complement of x, y; it is also a complex basis of E. In the ordered basis $z_0, e_1, \ldots, e_{n-2}$ the metric on T is

$$G = \operatorname{diag}(\kappa, 1, \ldots, 1), \tag{3.6}$$

and the ambient metric in (3.4) is $\operatorname{diag}(G, c)$.

The irreducible polynomial h defines a reduced analytic hypersurface, including locally. For example, a repeated local analytic factor would force all polynomial partial derivatives of h to vanish on an analytic set of dimension $n-1$, whereas the singular locus of an irreducible hypersurface in characteristic zero has dimension at most $n-2$. The divisibility $h \mid \mathcal{L}h$ makes $\mathcal{L}h$ vanish on X. Lemma 2.2 therefore applies and gives $\Delta_G \mu(0) = 0$.

Homogeneity also implies $\mu(az_0) = 0$ for all sufficiently small $a \in \mathbb{C}$. In fact,

$$h((1+a)z_0 + wN) = (1+a)^{\deg h}\, h\left(z_0 + \frac{w}{1+a}N\right).$$

In a sufficiently small fixed normal disk, the expression on the right has only the root zero, of multiplicity s. Thus all local roots on $\zeta = az_0$ vanish. If $\mu_{jj} = D^2\mu(0)[e_j, e_j]$, equation (3.6) consequently yields

$$\sum_{j=1}^{n-2} \mu_{jj} = 0. \tag{3.7}$$

A strictly negative trace. Fix j and put $\zeta = te_j$, where $t \in \mathbb{C}$. Orthogonality gives the exact identities

$$q(z) = \kappa + t^2 + cw^2,$$
$$|z|^2 = 1 + |t|^2 + 2(1-\kappa^2)\operatorname{Re} w + |N|^2|w|^2.$$

Use (3.5) to expand $\Phi(z) \le \log \kappa$, then average over the s local roots. We obtain

$$0 \ge \operatorname{Re}(t^2/\kappa) - |t|^2 - 2(1-\kappa^2)\operatorname{Re}\mu(te_j) + O(|t|^4).$$

Since $\mu(te_j) = \frac{1}{2}\mu_{jj}t^2 + O(|t|^3)$ and $c = -\kappa(1-\kappa^2)$, this reads

$$0 \ge \operatorname{Re}\left(\frac{1 + c\mu_{jj}}{\kappa}\, t^2\right) - |t|^2 + o(|t|^2). \tag{3.8}$$

Choose the phase of t to maximize the real part and let $|t| \to 0$. It follows that

$$|1 + c\mu_{jj}| \le \kappa, \qquad \operatorname{Re}(c\mu_{jj}) \le \kappa - 1 < 0.$$

Summing over j contradicts (3.7), since $n-2 \ge 1$. Therefore $\kappa > 0$ is impossible, and the theorem is proved. □

Remark 3.2. The argument uses the local Weierstrass preparation theorem, the squarefreeness of a reduced Weierstrass equation, and the complex Nullstellensatz; background on the analytic ingredients may be found in [1, Chapter II]. The proof above includes the tangent-cone and sheet estimates needed at a singular maximizing point. The assumption $n \ge 3$ is essential: in two variables a complex line may avoid every nonzero real point.

4. Homogeneous factors

Lemma 4.1 (Inheritance by an irreducible factor)**.** *Let $h \in \mathbb{C}[z_1, \dots, z_n]$ be irreducible and write $F = h^a v$, where $a \ge 1$ and $h \nmid v$. If $\mathcal{L}F = 0$, then $h \mid \mathcal{L}h$.*

Proof. There is a polynomial R such that

$$\mathcal{L}(h^a v) = h^{3a-3}\big(a^3 v^3 \mathcal{L}h + hR\big). \tag{4.1}$$

To verify this, write $g = \nabla h$, $b = \nabla v$, $U = D^2 h$, and $V = D^2 v$. Then $\nabla(h^a v) = h^{a-1}A$, where $A = avg + hb$, and formally $D^2(h^a v) = h^{a-2}B$, where

$$B = a(a-1)vgg^{\mathsf{T}} + h\{avU + a(gb^{\mathsf{T}} + bg^{\mathsf{T}})\} + h^2 V.$$

In $(A\cdot A)\operatorname{tr} B - A^{\mathsf{T}}BA$, the constant term in the displayed powers of h cancels, and the coefficient of h is $a^3v^3\mathcal{L}h$; the terms involving b cancel. Restoring the powers of h gives (4.1). For $a=1$, the potentially negative power in the formal Hessian formula disappears because $a(a-1)=0$.

Since the polynomial ring is a domain, cancel h^{3a-3} from (4.1). Reduction modulo the prime polynomial h gives $a^3v^3\mathcal{L}h \equiv 0 \pmod{h}$. As $h \nmid v$ and the characteristic is zero, we conclude that $h \mid \mathcal{L}h$. □

Lemma 4.2 (Exclusion of the radial factor)**.** *Let $n \geq 3$ and let $F \in \mathbb{C}[z_1,\dots,z_n]$ be nonzero and homogeneous. If $\mathcal{L}F = 0$, then $q \nmid F$.*

Proof. Otherwise, write $F = q^a g$, with $a \geq 1$ maximal, g homogeneous of degree $d \geq 0$, and $q \nmid g$. Product differentiation and Euler's identities give

$$\begin{aligned}\mathcal{L}(q^a g) = q^{3a-2}\big[&q^2\mathcal{L}g + 2a(n-2d-1-2a)qg(\nabla g\cdot\nabla g)\\ &+ 4a(a+d)qg^2\Delta g + 4a(a+d)\{2a(n-1)+d\}g^3\big].\end{aligned} \tag{4.2}$$

For completeness, the ingredients of this identity are

$$\begin{aligned}\nabla(q^a g) &= q^a\nabla g + 2aq^{a-1}gz,\\ \Delta(q^a g) &= q^{a-1}\{q\Delta g + 2a(n+2a-2+2d)g\},\\ D^2(q^a g) &= q^a D^2 g + 2aq^{a-1}\{z(\nabla g)^{\mathsf{T}} + (\nabla g)z^{\mathsf{T}} + gI\}\\ &\quad + 4a(a-1)q^{a-2}gzz^{\mathsf{T}},\end{aligned}$$

together with $z\cdot\nabla g = dg$ and $D^2 g\, z = (d-1)\nabla g$. For $a=1$ the last Hessian term vanishes.

Cancel q^{3a-2} from (4.2) and reduce modulo q. The equation $\mathcal{L}F = 0$ implies

$$4a(a+d)\{2a(n-1)+d\}g^3 \equiv 0 \pmod{q}.$$

The scalar coefficient is a nonzero positive integer. Since q is irreducible over $\mathbb{C}$ for $n \geq 3$, it follows that $q \mid g$, contrary to the choice of a. □

Corollary 4.3 (No positive homogeneous divisor)**.** *Let $n \geq 3$ and let $F \in \mathbb{R}[x_1,\dots,x_n]$ be nonzero and homogeneous, with $\mathcal{L}F = 0$. There is no nonconstant homogeneous real polynomial Q dividing F such that $Q(x) > 0$ for all $x \in \mathbb{R}^n \setminus \{0\}$.*

Proof. Choose a nonconstant irreducible complex factor h of Q. Factors of a homogeneous polynomial are homogeneous. The positivity of Q implies that h has no nonzero real zero. Since h also divides F, Lemma 4.1 gives $h \mid \mathcal{L}h$, regardless of its multiplicity in F. Theorem 3.1 gives $h = \alpha q$, so $q \mid F$. This contradicts Lemma 4.2. □

5. Completion in dimension eight

We state precisely the two results from Guo that are needed here. In the terminology of [2], a polynomial solution is nonaffine; additive constants do not affect the equation.

Proposition 5.1 (Guo's structure and degree theorems)**.** *Let $n \geq 8$, let P be a nonaffine polynomial solution of (1.1) on $\mathbb{R}^n$, and let $F = P_m$ be its highest homogeneous component. Then*

$$F = p^k Q,$$

where p is an irreducible homogeneous real polynomial, $k \geq 1$ is odd, and Q is a nonzero nonnegative homogeneous real polynomial. Moreover,

$$\dim\{Q = 0\} \leq n - 8 \tag{5.1}$$

and

$$\mu_n^- < \deg p + \frac{\deg Q}{k} < \mu_n^+, \qquad \mu_n^{\pm} = \frac{n-1 \pm \sqrt{(n-3)^2 - 4(n-2)}}{2}. \tag{5.2}$$

Source. The factorization and (5.1) are the portions of [2, Theorem 1.3(ii)] used here. The strict inequality (5.2) is [2, Theorem 1.5]. In the body of that paper these are Theorems 4.7(ii) and 5.2, respectively. □

Proof of Theorem 1.1. Suppose P has degree $m \geq 2$, and let $F = P_m \neq 0$. The homogeneous component of degree $3m-4$ in $\mathcal{M}P = \Delta P + \mathcal{L}P$ is $\mathcal{L}F$. Indeed, second derivatives have degree at most $m-2$, and the terms of degree $3m-4$ in the cubic differential expression involve only P_m. Since $3m-4 > m-2$, it follows that $\mathcal{L}F = 0$.

Apply Proposition 5.1 with $n = 8$ to write $F = p^k Q$. If Q were nonconstant and vanished at some nonzero real point, homogeneity would place an entire nonzero real ray in $\{Q = 0\}$. This would contradict $\dim\{Q = 0\} \leq 0$. Since Q is nonnegative, it would therefore be strictly positive on $\mathbb{R}^8 \setminus \{0\}$. Corollary 4.3 excludes such a nonconstant divisor. Thus Q is constant and $\deg Q = 0$.

In dimension eight, the endpoints in (5.2) are

$$\mu_8^- = \frac{7 - \sqrt{25-24}}{2} = 3, \qquad \mu_8^+ = \frac{7 + \sqrt{25-24}}{2} = 4.$$

Consequently $3 < \deg p < 4$, which is impossible for an integer degree. Therefore $\deg P \leq 1$, giving the stated affine form. Conversely, every affine polynomial has zero Hessian and satisfies (1.1). □

6. Companion Lean proof

The accompanying file `Math30Source29.lean` [3] proves Theorem 1.1 as the exact catalogue statement `Math30Catalog.source29`. Its terminal theorem is

```
theorem target : fcTypeOfName% "Math30Catalog.source29" := by
  exact Source29FiniteDimensionalRigidity.problem29
```

The file is a submission body. Its catalogue environment supplies the imports

```
import FormalConjectures.ResearchTargets.Math30
import TaskSupport
```

and the surrounding `Bounty` namespace. These are required to interpret the catalogue statement and its definitions.

The formal proof reaches the same theorem by a longer geometric route. In particular, it develops calibrated homogeneous rigidity and derives phase charts by induction in the dimension. The written proof in Sections 2–5 and the formal development thus follow different routes to the same statement. The principal final steps are as follows:

(1) The leading-component reduction proves that a nonaffine polynomial solution has a nonzero homogeneous leading form annihilated by $\mathcal{L}$. The formal development also derives the weak calibration of this form.
(2) Starting from the two-dimensional homogeneous case, regular-cone induction proves `linearRigidity_seven`.
(3) This supplies `phase_chart_dimension_eight` at every nonzero zero of the leading form.
(4) A proved reduction combines these charts with the remaining structure and degree arguments to obtain `problem29`.

The last three named declarations are in the namespace

```
Bounty.Source29FiniteDimensionalRigidity
```

The retained local verification record for the supplied submission reports acceptance of the exact target, equality with the catalogue statement, and successful Lean kernel checking. Its permitted axioms are `propext`, `Quot.sound`, and `Classical.choice`; no additional mathematical axioms or proof placeholders occur in the submitted body. An independent local audit on October 9, 2026 also checked the type of the compiled target against the explicit implication $\mathcal{M}P = 0 \Rightarrow \deg P \leq 1$ for eight variables and confirmed this axiom list. These are local checks in the pinned catalogue environment. The companion file records a formal proof of the theorem; the complex-analytic presentation above uses the explicitly cited results of Guo.

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